

Year 7

Year 8

IGCSE

A-Level

I.B

ICT Fun

Forum

Homepage

Section 3.6: Solid State Storage Media

In this section you will see the 4 different types of solid state storage media.

- Solid State Hard Drive
- Memory stick
- Flash memory
- Micro flash memory

Key Concepts of this section:

- # Know the **different types** of solid state storage media.
- # Understand the **uses of** each solid state media.
- # Be able to **discuss advantages and disadvantages** of each solid state media.

For each we will discuss **features, uses, advantages/disadvantages**.

Solid State Media (an overview)

Definition: "Any memory device with no moving parts and are based on electronic circuits."

Overview of Solid State Media:

- # Solid State Drives have **no moving parts**.

No reels of tape, no spinning disks, no moving laser beams etc.

- # SSD technology is known as '**flash memory**' and this is the **smallest** form of data storage.

Examples include **memory sticks, memory cards** etc.

- # They store data as 1's and 0's within **millions of electrical transistors** instead of on films of magnetic substance or as dots on discs.

If the transistor **conducts an electric current**, this equates a **1**.

If it **does not conduct a current**, this equates a **0**

Animation showing how flash memory works.

- # They hold several advantages over Magnetic Disc Drives:

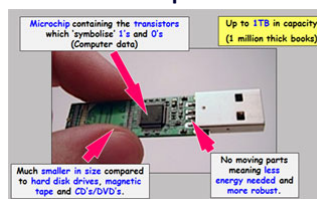
- **Much smaller**
- **Use less power** (no moving parts means less power needed)
- **Faster data access and transfer times**
- **Much more robust** (harder to damage)



Key Words:

Flash memory, No moving parts, Electronic circuits

Examples:



Summary of solid state drives.
(Click to zoom)



Solid state hard drives are much smaller than magnetic hard disks.

NOTE:

All the examples of solid state storage media on this page use **direct data access**.

Examples of Solid State Media

- # The table below shows you some examples of solid state storage media. Click each of the media to move to more information:

Solid State Hard Drives



(Direct Access)

Memory Stick / Pen Drive



(Direct Access)

Flash Memory Cards



(Direct Access)

Micro Flash Memory Cards



(Direct Access)

REMEMBER!

Solid state media have **no moving parts**.

NOTE:

Click the images to the left to move to more information about each one.

Breakdown of each output device:

- # The following sections will help summarise each of the manual input devices. For each one you will see:

- **Information about the media**

- **Uses of the media**
- **Advantages of the media**
- **Disadvantages of the media**

1 Solid State Hard Drives (SSD's)

Information:

- # Used in **desktop PC's** and **laptops** in place of magnetic hard disk drives.
- # They are also known as '**SSD's**'.
- # They offer advantages over magnetic hard drives such as:
 - **Run silently** (magnetic hard disk drives spin producing whirring sound)
 - **Longer life span**
 - **Faster data access and transfer times**
 - **Use less power**
 - **Produce less heat**
- # They use **solid state technology** based on electrical circuits which means they have **no moving parts**.
No moving parts = less chance of them breaking down.

Magnetic hard disks have **moving parts** such as spinning disks and read/write heads.

Eventually these parts **stop functioning**.

- # These store files **permanently** (non-volatile storage).
- # They have a current **capacity** of about **1TB** (approx 1 million thick books worth of text)

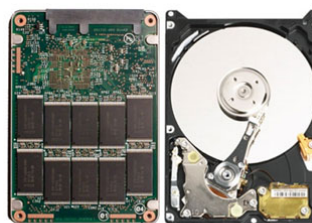
Uses of Solid State Hard Drives:

- # Used in desktop PC's and laptops to permanently store data such as:
 - **Operating systems**
 - **Software applications** (such as word processors, spreadsheets etc)
 - **Music/Video files**
 - **School work, photos**

Examples:



SSD's are used in place of magnetic hard disks in laptops and PC's.



Solid State Drive

Data stored in electrical circuits with no moving parts

Magnetic Hard Disk

Data stored on a spinning platter and read/write head

SSD's compared to magnetic hard disk drives.
(Click to zoom)

Advantages/Disadvantages of Solid State Hard Drives:

Advantages	Disadvantages
Last longer than magnetic hard disks (no moving parts).	Smaller storage capacity than magnetic hard disks.
Faster data access/transfer times. Note: Computers even boot up faster using solid state hard drives.	More expensive than magnetic hard disks.
Use less power and produce less heat.	
Run silently .	

2 Memory Sticks / Pen Drives

Information:

- # Memory sticks are **small, portable external** storage devices.
- # They are used to **easily transfer** data between computers.
- # Memory sticks use **solid state technology** and are usually **connected to the computer** via **USB** ports.

Note: Memory sticks are making other forms of portable storage (like CD's and DVD's) **redundant** as they are **much easier** and **quicker** to **transfer data**.

- # **Data access** and **transfer** is very fast.
- # Memory sticks are **very cheap** to buy.

You can currently get 32GB of storage for less than £10!

- # Current **capacities** of memory sticks are up to **512GB** (about 500,000 thick books)

Uses of Memory Sticks:

- # Used for easily **transporting files/data** between computers.
- # Used for **backing up data** quickly and easily.

Examples:



Summary of memory sticks.
(Click to zoom)



Memory sticks are very small and connect to computers via USB.

Advantages/Disadvantages of Memory Sticks:

Advantages	Disadvantages
Very small and lightweight (portable).	No write-protect feature . This means that it is possible to accidentally copy over data
Robust - no moving parts means they are less likely to become damaged. For example: CD's and DVD's can get scratched.	Small physical size means that they are easy to misplace or lose
Compatible with almost any computer. (most computers have USB ports)	More expensive per GB than magnetic hard disks.
Large storage capacities (up to 512 GB)	



Memory sticks have large capacities.

3 Flash Memory Cards (and Micro Flash Memory Cards)

Information:

- # Flash memory cards are also known as '**Secure Digital Cards**' (or SD Cards).
- # SD cards are **small and rectangular** in shape.
- # These are a form of **Electronically Erasable Programmable Read Only Memory** (EEPROM).
- # SD cards use **solid state technology** (they have no moving parts).

Note: **Micro flash memory cards** are just smaller versions of the regular flash memory cards.

Micro flash memory cards are sometimes also called '**Micro SD Cards**'.

- # A wide range of devices have slots to read flash memory cards including **laptops, mobile phones, tablets** and **mp3 players**.
- # SD card readers can be used to transfer data from the cards onto a desktop PC.
- # **SD Cards** have a maximum **capacity** of about **256GB**
- # **Micro SD Cards** have a maximum **capacity** of about **128GB**

Uses of Flash Memory Cards:

- # SD cards are used to **store photos** on digital cameras.

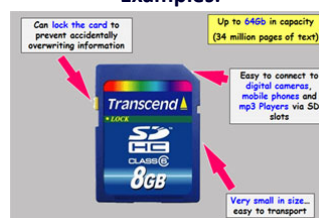
Note: Many **stores** that offer **printing services** have **SD card readers** that access data on your cards.

This means you can get your photographs printed without even having to take your digital camera with you to the store.

- # SD cards are used in **hand held devices** (like PDA's and palmtop's) to store **files** and **data**.
- # Micro SD cards are used in **mobile phones as memory cards** to store **photos** and **music** etc.
- # Micro SD cards can be used by **MP3 players** to store **music** files

Advantages/Disadvantages of Flash Memory Cards:

Advantages	Disadvantages
Very small so they are easy to transport files from one device to another (Camera to Camera for example).	More expensive per GB than magnetic hard disk drives.
Robust and not easily damaged (No moving parts).	Lower storage capacity than magnetic hard disks.

Examples:Summary of memory sticks.
(Click to zoom)

SD cards and Micro SD cards are really the same thing except for the size.



Micro SD cards have a very small physical size.



Flash memory is used as storage in devices such as mp3 players.

Easy to connect to mobile devices through SD Slots

Small size makes them easy to have stolen or lose.

Desktop PC's don't usually come with built-in memory card readers. To connect to a desktop PC you need to buy an **SD card reader**.

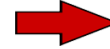


Flash card readers can connect flash cards to PC's



[Previous - Optical Storage Media](#)

[Next - Comparison of Storage Media](#)



Links to Theory Units:

[Section 1: Types and Components of Computer Systems](#)

[Section 2: Input and Output Devices](#)

[Section 3: Storage Devices and Media](#)

[Section 4: Computer Networks](#)

[Section 5: Data Types](#)

[Section 6: The Effects of Using ICT](#)

[Section 7: The Ways in which ICT is used](#)

[Section 7: Part 2](#)

[Section 8: Systems Analysis and Design](#)

Links to Practical Units:

[Section 9: Communication](#)

[Section 10: Document Production](#)

[Section 11: Data Manipulation](#)

[Section 12: Integration](#)

[Section 13: Output Data](#)

[Section 14: Data Analysis](#)

[Section 15: Web site Authoring](#)

[Section 16: Presentation Authoring](#)

[Back to top](#)